Quality-Aware Memory Gating for Multi-Modal Maritime Segmentation with Foundation Models

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Abstract

Multi-modal semantic segmentation from unmanned surface vehicles must cope with severe sensor degradation—particularly RGB cameras at night—while exploiting complementary thermal and LiDAR information. MemorySAM recently showed that SAM2's temporal memory mechanism can be repurposed for cross-modal fusion by treating each modality as a sequential frame. However, this sequential design has a structural vulnerability: the first modality (typically RGB) initializes the memory bank, and all subsequent modalities attend to it via cross-attention. When RGB is degraded at night, its low-quality features pollute the shared memory, corrupting the cross-modal attention for thermal and LiDAR. We propose Quality-Aware Memory Gating *to address this memory pollution. During training, a frozen copy of SAM2's decoder independently segments each modality to produce per-pixel cross-entropy maps against the ground truth, yielding spatial quality targets. A lightweight convolutional gating network (∼12.5K parameters) learns to predict these quality maps from the encoded features alone, and at inference the predicted maps spatially modulate memory features before they enter the bank—preserving high-quality regions while suppressing degraded ones—without additional decoding cost. Combined with Soft Mixture-of-Experts LoRA for modality-adaptive encoding and aggressive night-simulation augmentations, we validate our approach on the MULTIAQUA benchmark for daytime-to-nighttime maritime segmentation.*

1. Introduction

Reliable scene understanding is a critical capability for unmanned surface vehicles (USVs) operating in maritime environments. Unlike structured road scenes, waterways present unique challenges: water surfaces exhibit highly variable reflectance, obstacles appear at unpredictable scales and distances, and visibility conditions can shift dramatically between day and night. Multi-modal sensing—combining RGB cameras with thermal and LiDAR sensors—offers a principled approach to address these challenges, as each modality provides complementary information robust to different environmental degradations [22, 23].

However, effectively fusing heterogeneous modalities remains an open problem, especially when individual sensor quality varies across conditions. Traditional multi-modal segmentation methods [10, 16, 23] design modality-specific encoders or fusion modules, which scale poorly as the number of modalities increases. Meanwhile, foundation models such as SAM [9] and SAM2 [15] have demonstrated remarkable segmentation capabilities through large-scale pretraining, but they are fundamentally designed for single-modal RGB input and instance-level tasks.

MemorySAM [4] elegantly bridges this gap by reinterpreting SAM2's temporal memory mechanism for cross-modal aggregation: each modality is treated as a sequential "frame", and the memory bank enables rich cross-modal information exchange through SAM2's built-in cross-attention, achieving strong results on multi-modal benchmarks.

While this sequential memory design is effective in benign conditions, we observe that it becomes vulnerable when sensor quality is asymmetric. In MemorySAM's pipeline, the first modality to be processed initializes the memory bank, and all subsequent modalities attend to it via cross-attention. When the leading modality is reliable, this naturally bootstraps strong cross-modal representations. However, in nighttime maritime scenes—where RGB cameras capture near-zero signal while thermal and LiDAR remain informative—the degraded RGB features still enter the memory bank as the initial reference. Subsequent modalities then attend to this low-quality memory, and the noise propagates through cross-attention into what would otherwise be reliable thermal and LiDAR representations.

We term this phenomenon *memory pollution*: degraded features from one modality contaminate the shared memory bank and undermine cross-modal fusion for all subsequent modalities. Crucially, this problem cannot be resolved by adjusting modality weights at the output stage, because the shared memory representation is already corrupted before output fusion takes place.

To address memory pollution, we propose *Quality-Aware Memory Gating*, which intervenes directly at the memory storage stage. Our approach uses a teacher–student distillation framework:

- **Teacher** (training only): SAM2's decoder independently segments each modality without cross-modal memory, producing a per-pixel cross-entropy map against the ground truth. This reveals the spatial regions where each modality makes accurate predictions versus where it fails—capturing structural limitations such as LiDAR's inability to sense sky or RGB's failure in darkness.
- **Student** (training and inference): A lightweight convolutional network ($\sim$12.5K parameters) learns to predict this spatial quality map from the encoded features alone. The predicted map multiplicatively gates memory features before they enter the bank, suppressing degraded regions while preserving reliable ones.

At inference, only the student network runs, adding negligible computational cost while preventing degraded features from polluting the shared memory.

Our full pipeline builds on the MemorySAM framework and additionally incorporates: **(1)** Soft Mixture-of-Experts (Soft-MoE) gating [14] in SAM2's LoRA [7] layers for modality-adaptive feature extraction with full expert utilization; **(2)** a cross-modal fusion head for relative modality weighting applied through dual paths; and **(3)** physics-inspired RGB augmentations [12, 19] for day-to-night generalization.

We evaluate on the MULTIAQUA benchmark [12], which provides synchronized RGB, thermal, and LiDAR data from USVs across daytime and nighttime conditions. The remainder of this paper is organized as follows: Sec. 2 reviews related work, Sec. 3 details our method, and Sec. 4 presents experimental results.

2. Related Work

2.1. Multi-Modal Semantic Segmentation

Multi-modal semantic segmentation leverages complementary information from heterogeneous sensors to improve scene understanding. Early dual-encoder approaches [5, 20] fuse fixed modal pairs via element-wise operations, but do not generalize to arbitrary modality combinations. Transformer-based methods have advanced this field considerably. CMX [22] introduced cross-modal feature rectification for long-range context exchange, and CMNeXt [23] scaled this to arbitrary numbers of modalities with negligible per-modality overhead. StitchFusion [10] repurposes pretrained encoders as feature fusers, while MMSFormer [16] progressively integrates modalities within a unified Transformer block. Channel exchanging [21] dynamically swaps feature channels across modality branches. For modality robustness, Complementary Random Masking (CRM) [19] randomly masks input modalities during training to prevent over-reliance on any single sensor. Our night augmentation pipeline extends this idea with extreme brightness corruption and full-channel zeroing for maritime nighttime scenarios.

2.2. Maritime Scene Understanding

Maritime perception differs from urban driving in its highly reflective water surfaces, sparse obstacles, and dramatic illumination changes. Benchmarks such as LaRS [26] and MaSTr1325 [1] have driven progress in RGB-based maritime obstacle detection, while the MACVi workshop [2, 8] provides standardized evaluation protocols. The MULTIAQUA dataset [12] extends maritime perception to the multi-modal setting with synchronized RGB, thermal, and LiDAR data, and introduces a challenging domain shift where models trained on daytime images must generalize to nighttime conditions with severely degraded RGB.

2.3. Foundation Model Adaptation for Segmentation

The Segment Anything Model (SAM) [9] established a new paradigm for promptable segmentation through large-scale pretraining, and SAM2 [15] extended it to video with a streaming memory architecture built on the Hiera backbone [17]. The strong generalization of SAM has motivated its adaptation to diverse domains: medical imaging [11], remote sensing [3], 3D point clouds [24], and robotic grasping [13], demonstrating that SAM's pretrained representations transfer effectively with appropriate domain-specific strategies.

For parameter-efficient adaptation, LoRA [7] injects trainable low-rank matrices into frozen attention layers. MoE-LoRA [25] extends this with Mixture-of-Experts [18] routing for modality-adaptive feature extraction, though hard top-k selection can underutilize experts when the expert count is small. Soft MoE [14] addresses this by replacing discrete routing with continuous softmax weighting.

MemorySAM [4] repurposes SAM2's temporal memory mechanism for multi-modal fusion by treating each modality as a sequential frame, enabling cross-modal feature exchange through the memory bank without explicit fusion modules. A Semantic Prototype Memory Module further bridges the gap from instance-level to semantic-level understanding. Our work builds on this framework and addresses a limitation of its sequential design: when sensor quality is asymmetric (*e.g.*, degraded RGB at night), low-quality features can pollute the shared memory and corrupt cross-modal attention for subsequent modalities. We propose quality-aware memory gating, using a teacher–student distillation [6] scheme to spatially modulate memory features before they enter the bank.

3. Method

Our method builds on MemorySAM [4], which adapts SAM2 [15] for multi-modal semantic segmentation by treating each modality as a sequential frame processed through the memory-based tracking pipeline. We retain this sequential processing paradigm and introduce quality-aware memory gating to address the memory pollution problem. Our pipeline operates in five phases: given M input modalities (RGB, thermal, LiDAR), (1) each modality is encoded by SAM2's frozen backbone with Soft-MoE LoRA layers; (2) a cross-modal fusion head produces relative modality weights; (3) the modality features are modulated by uncertainty-aware weights before entering the tracking step; (4) a spatial quality gating network modulates the resulting memory features before they are stored in the memory bank; and (5) the per-modality outputs are fused using the relative weights.

3.1. Soft-MoE LoRA

We keep SAM2's image encoder frozen and inject low-rank adaptation matrices into the query and value projections of each attention block, following the standard LoRA [7] formulation. Rather than using a single pair of low-rank matrices, we maintain N expert pairs $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{N}$, where $\mathbf{A}_i \in \mathbb{R}^{d\times r}$ and $\mathbf{B}_i \in \mathbb{R}^{r\times d}$, with r being the LoRA rank and d the feature dimension.

Following the Soft Mixture-of-Experts formulation [14], instead of selecting the top-k experts via hard routing [18, 25], we compute a softmax-weighted combination over all experts for each input token $\mathbf{x}$:

$$\Delta\mathbf{x} = \sum_{i=1}^{N} \alpha_i(\mathbf{x}) \cdot \mathbf{B}_i\mathbf{A}_i\mathbf{x}, \quad \alpha_i(\mathbf{x}) = \frac{\exp(g_i(\mathbf{x}))}{\sum_{j=1}^{N}\exp(g_j(\mathbf{x}))}, \tag{1}$$

where $g(\mathbf{x}) = \mathbf{W}_g\mathbf{x} + \mathbf{b}_g$ is a lightweight gating network with $\mathbf{W}_g \in \mathbb{R}^{N\times d}$. The output of the original QKV projection is augmented as $\mathbf{q} \leftarrow \mathbf{q} + \Delta\mathbf{q}$ and $\mathbf{v} \leftarrow \mathbf{v} + \Delta\mathbf{v}$. When the number of experts N is small (*e.g.*, N=3), hard top-k routing with k=2 permanently disables one expert per token; soft gating avoids this by ensuring gradient flow to every expert at every step. We use N=3 experts and r=4 throughout our experiments.

3.2. Cross-Modal Fusion with Dual-Path Weighting

Cross-modal scoring. In addition to the memory-level quality gating described in Sec. 3.3, we employ a cross-modal fusion head for output-level modality weighting. Rather than scoring each modality independently, which can lead to degenerate solutions when all scores collapse to similar values, we jointly compare all modality features to produce relative weights. Given backbone feature maps $\{\mathbf{F}_i\}_{i=1}^{M}$, each is compressed by a shared network into a compact vector:

$$\mathbf{z}_i = \mathrm{ReLU}(\mathbf{W}_c \cdot \mathrm{GAP}(\mathbf{F}_i) + \mathbf{b}_c), \tag{2}$$

where GAP denotes global average pooling and $\mathbf{W}_c \in \mathbb{R}^{h\times C}$ are shared weights. The compressed vectors are concatenated and passed through a comparison MLP to produce modality weights:

$$\mathbf{l} = \psi([\mathbf{z}_1; \mathbf{z}_2; \ldots; \mathbf{z}_M]), \quad w_i = \frac{\exp(l_i/\tau)}{\sum_{j=1}^{M}\exp(l_j/\tau)}, \tag{3}$$

where ψ is a two-layer MLP and τ is a temperature parameter. We initialize the output layer of ψ to zero, ensuring uniform weights $1/M$ at the start of training for stable initialization.

Uncertainty-Aware Memory Modulation (UAMM). Before each modality's features enter SAM2's memory-augmented tracking step, we modulate them using max-normalized weights:

$$\hat{w}_i = \frac{w_i}{\max_j w_j + \epsilon}, \quad \hat{\mathbf{f}}_i^{(k)} = \hat{w}_i \cdot \mathbf{f}_i^{(k)}, \quad \forall k, \tag{4}$$

where $\mathbf{f}_i^{(k)}$ are the multi-scale vision features for modality i and $\epsilon = 10^{-8}$. The max-normalization ensures that the highest-scoring modality always receives $\hat{w} = 1.0$ (no attenuation), while other modalities are scaled down proportionally.

The modulated features are then processed by SAM2's tracking step, where the first modality initializes the memory and subsequent modalities attend to it via cross-attention, enabling cross-modal feature exchange through SAM2's built-in memory attention mechanism.

Adaptive Mask Fusion (AMF). After obtaining per-modality mask predictions $\mathbf{o}_i$ from the tracking loop, we fuse them using the raw softmax weights:

$$\mathbf{o}_{\text{fused}} = \sum_{i=1}^{M} w_i \cdot \mathbf{o}_i, \tag{5}$$

where w_i are the softmax weights from Eq. (3). This dual-path design—max-normalized for UAMM, raw softmax for AMF—serves complementary purposes: UAMM requires absolute preservation of the best modality's feature magnitude, while AMF requires relative weighting proportional to estimated quality.

3.3. Quality-Aware Memory Gating

The cross-modal fusion head and UAMM modulate the vision features *before* the tracking step, but the memory features generated *after* the tracking step are stored in the memory bank without any quality-based filtering. When a degraded modality (*e.g.*, RGB at night) produces low-quality memory features, subsequent modalities attend to these corrupted features through SAM2's cross-attention mechanism, propagating errors across the entire fusion pipeline. We address this by introducing a spatial quality gating module that modulates memory features before they enter the bank.

3.3.1. Teacher: Per-Modality Decoder Quality Assessment

During training, we generate spatial quality targets by independently decoding each modality's features through SAM2's decoder *without* cross-modal memory. For each modality m, the encoded vision features $\mathbf{f}_m$ are passed through SAM2's mask decoder in single-frame mode, bypassing the memory bank entirely:

$$\hat{\mathbf{y}}_m = \mathrm{Decoder}(\mathbf{f}_m; \emptyset), \tag{6}$$

where $\emptyset$ denotes an empty memory bank and $\hat{\mathbf{y}}_m \in \mathbb{R}^{B\times C\times H\times W}$ are the per-modality logits with C semantic classes. The per-pixel cross-entropy against the ground truth $\mathbf{y}^*$ provides a spatial error map:

$$\mathcal{E}_m(p) = -\sum_{c=1}^{C} \mathbf{y}^*_c(p) \log \hat{\mathbf{y}}_{m,c}(p), \tag{7}$$

where p indexes spatial positions. This error map is converted into a quality target:

$$\mathbf{q}^*_m = \mathrm{downsample}\big(\exp(-\mathcal{E}_m)\big) \in [0,1]^{H'\times W'}, \tag{8}$$

where $H' \times W'$ matches the feature map resolution. Positions where the modality produces accurate predictions yield quality values near 1.0, while positions with high error yield values near 0.

This teacher signal has two important properties compared to confidence-based alternatives: **(i)** It measures *correctness* against the ground truth, not merely confidence—a modality that is "confidently wrong" (*e.g.*, LiDAR predicting water where there is sky) receives low quality. **(ii)** It captures *structural* modality limitations that persist even as the model converges: LiDAR inherently cannot distinguish sky from water, so its quality target for sky regions remains low throughout training, providing a non-diminishing supervision signal.

The teacher decoder is frozen during this process and adds no gradient computation cost.

3.3.2. Student: Spatial Quality Gating Network

A lightweight convolutional network predicts the spatial quality map from the encoded features:

$$\hat{\mathbf{q}}_m = \sigma(\mathcal{G}(\mathbf{f}^{(0)}_m)) \cdot (1 - q_{\min}) + q_{\min}, \tag{9}$$

where $\mathbf{f}^{(0)}_m$ is the highest-resolution FPN feature, σ is the sigmoid function, $q_{\min} = 0.1$ prevents complete zeroing of memory features, and $\mathcal{G}$ is a three-layer CNN:

$$\mathcal{G}: \mathrm{Conv}_{3\times3}(C_{\text{in}}, 64) \to \mathrm{ReLU} \to \mathrm{Conv}_{3\times3}(64, 64) \to \mathrm{ReLU} \to \mathrm{Conv}_1 \tag{10}$$

The final layer's bias is initialized to $+1.0$, yielding $\sigma(1.0) \approx 0.73$, so that all memory features pass through with moderate gating at the start of training (optimistic initialization). The network contains ∼12.5K parameters.

3.3.3. Memory Modulation

After each modality's tracking step, SAM2's memory encoder produces memory features $\mathbf{m}_i$ to be stored in the bank. We apply the predicted quality map before storage:

$$\tilde{\mathbf{m}}_i = \mathbf{m}_i \odot \mathrm{resize}(\hat{\mathbf{q}}_i), \tag{11}$$

where $\odot$ denotes element-wise multiplication with spatial broadcasting, and resize adjusts the quality map to match the memory feature resolution via bilinear interpolation. The gated memory $\tilde{\mathbf{m}}_i$ is then stored in the memory bank for subsequent modalities to attend to.

This design has a direct effect on the cross-modal attention of later modalities. Consider a nighttime scene where RGB is processed first (frame index 0): the quality gating network suppresses degraded RGB regions (*e.g.*, sky and water where RGB captures near-zero values), so when LiDAR (frame index 1) and thermal (frame index 2) attend to the memory bank, they receive attenuated contributions from RGB's weak regions while retaining information from regions where RGB remains informative. Conversely, LiDAR's quality map suppresses sky regions (where LiDAR has no information) but preserves object regions (where LiDAR excels), allowing thermal to selectively leverage LiDAR's strengths.

3.4. Training Objectives

Following MemorySAM [4], we use a Semantic Prototype Memory Module: for each class c, a global prototype $\mathbf{p}_c$ is maintained via exponential moving average (momentum $= 0.8$), and the prototype loss is $\mathcal{L}_{\text{proto}} = \mathrm{MSE}(\hat{\mathbf{p}}, \mathbf{p})$. The quality gating loss is the binary cross-entropy between the predicted quality logits and the teacher target:

$$\mathcal{L}_{\text{gate}} = \frac{1}{M}\sum_{m=1}^{M} \mathrm{BCE}(\mathcal{G}(\mathbf{f}^{(0)}_m),\ \mathbf{q}^*_m), \tag{12}$$

where ignore-class pixels (label 255) are excluded from the loss computation. The total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{OHEM}} + \mathcal{L}_{\text{proto}} + \lambda_{\text{gate}}\mathcal{L}_{\text{gate}}, \tag{13}$$

where $\mathcal{L}_{\text{OHEM}}$ is the Online Hard Example Mining cross-entropy loss and λ_{gate} is a weighting coefficient. The cross-modal fusion head is trained end-to-end through the segmentation objective without additional supervision. The teacher decoder is frozen, so the quality gating loss gradient flows only through the student gating network.

4. Experiments

4.1. Dataset and Setup

Dataset. We evaluate on the MULTIAQUA dataset [12], which provides synchronized RGB, thermal, and LiDAR data captured from unmanned surface vehicles. The dataset contains four semantic classes: static obstacle, dynamic obstacle, water, and sky. The training and validation sets consist of daytime images (145 validation images), while the test set contains only nighttime images evaluated through the MACVi challenge server [2]. This setup requires models trained on daytime data to generalize to nighttime conditions where RGB quality degrades severely.

Implementation details. We use SAM2 [15] with a Hiera-B+ [17] backbone as the foundation model, with LoRA [7] rank $r{=}4$ and $N{=}3$ Soft-MoE experts. All images are resized to 1024×1024. We train with AdamW optimizer (learning rate $6{\times}10^{-4}$, weight decay 0.01) using a warmup-polynomial schedule (10 warmup epochs, power 0.9). The batch size is 1 per GPU and we use Online Hard Example Mining (OHEM) cross-entropy loss with the prototype segmentation loss [4] (Sec. 3.4). SAM2's image encoder is frozen; only the Soft-MoE LoRA layers, cross-modal fusion head, quality gating network, prompt encoder, and mask decoder are trained. The quality gating loss weight is $\lambda_{\text{gate}} = 0.5$. To bridge the day-to-night domain gap, we apply physics-inspired RGB augmentations during training [12]: brightness and gamma reduction simulating low-light conditions, complementary random masking [19], and stochastic full-channel zeroing, while leaving thermal and LiDAR unchanged. Training is conducted with PyTorch DDP on NVIDIA GPUs.

Evaluation metric. We report mean Intersection-over-Union (mIoU) averaged over the four semantic classes, on both the daytime validation set and the nighttime test set.

4.2. Comparison with State of the Art

Tab. 1 compares our method against prior approaches on the MULTIAQUA benchmark. The baselines include CMNeXt [23], MMSFormer [16], and StitchFusion [10] with various training strategies from [12]: suffix "-D" denotes double-pass training (processing each modality pair twice with swapped order), and "-H" denotes a multi-head decoder. All baseline results are taken from [12].

Tab. 1 shows that among the baselines, the double-pass training strategy ("-D") provides the largest improvement for nighttime generalization, with CMNeXt-DH achieving 74.25% test mIoU.

Table 1. Comparison on the MULTIAQUA benchmark. Val mIoU (daytime) and test mIoU (nighttime) are reported. "D" = double-pass training, "H" = multi-head decoder. Baseline results from [12]. Best results in **bold**, second best <u>underlined</u>.

Method	Val mIoU	Test mIoU	Obs. (test)
CMNeXt$_{\text{RGB}}$ [23]	**94.31**	38.23	11.87
CMNeXt [23]	88.78	50.52	10.86
CMNeXt-H	89.97	50.05	11.10
CMNeXt-D	92.95	72.24	32.68
CMNeXt-DH	93.58	<u>74.25</u>	<u>37.25</u>
MMSFormer [16]	85.68	40.11	10.37
MMSFormer-H	88.36	45.70	9.37
MMSFormer-D	89.98	64.12	13.36
MMSFormer-DH	88.21	63.09	15.89
StitchFusion [10]	93.57	41.86	14.98
StitchFusion-H	91.14	42.95	19.73
StitchFusion-D	89.81	74.23	36.89
StitchFusion-DH	91.59	73.59	34.38
MemorySAM [4]	**[TODO: XX.X]**	**[TODO: XX.X]**	**[TODO: XX.X]**
Ours	**[TODO: XX.X]**	**[TODO: XX.X]**	**[TODO: XX.X]**

Table 2. Ablation on night-robust RGB augmentation.

Night Aug	Val mIoU	Test mIoU
✗	93.10	35.93
✓	**[TODO: XX.X]**	**[TODO: XX.X]**

Table 3. Ablation on quality-aware memory gating. "QMG" = quality-aware memory gating.

QMG	Val mIoU	Test mIoU
✗(P9 baseline)	93.54	70.41
✓(Ours)	**[TODO: XX.X]**	**[TODO: XX.X]**

4.3. Ablation Studies

Effect of night augmentation. Tab. 2 validates the importance of night-robust augmentation. Without any RGB corruption, the model achieves high daytime validation mIoU but suffers catastrophic failure on the nighttime test set. Night augmentation substantially bridges this gap by exposing the model to simulated degradation patterns during training.

Effect of quality-aware memory gating. To isolate the contribution of quality-aware memory gating, we compare our full model against the baseline without the gating module (equivalent to the MemorySAM backbone with our other components).

5. Conclusion

We identified a structural vulnerability in memory-based multi-modal fusion with foundation models: when modalities are processed sequentially through SAM2's memory mechanism, degraded sensors (*e.g.*, RGB at night) pollute the shared memory bank and corrupt cross-modal attention for subsequent, otherwise reliable modalities. To address this, we proposed Quality-Aware Memory Gating, a teacher–student framework that spatially modulates memory features before they are stored. The teacher leverages SAM2's own decoder to assess per-modality spatial quality during training, while the lightweight student network (∼12.5K parameters) learns to predict quality maps at inference with negligible overhead. Combined with Soft-MoE LoRA for modality-adaptive encoding and night-simulation augmentations for domain generalization, our approach addresses the key challenge of multi-modal maritime segmentation under varying sensor conditions. Experiments on the MULTIAQUA benchmark validate our design choices.

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